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Growth on 2019 State Achievement Tests: Empirical Benchmarks and the Role of Scale Choice

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ABSTRACT
Benchmarks for "years of learning"–expectations for annual academic growth in standard deviation units–are popular to contextualize the practical significance of a causal effect. Though they are applied in studies using a variety of outcome measures, these benchmarks must be built using vertically scaled test scores. Vertical scales must be established to measure growth across grades, and different scales can lead to different conclusions about how much students grow. This article introduces new benchmarks for annual growth based on 2019 grade 3–8 state achievement tests in math and ELA, which are more appropriate for use in contemporary studies using test scores as an outcome measure than existing benchmarks based on older tests aligned to prior generation standards. While high-level trends in growth are generally consistent between prior studies and the present study, differences do arise that would lead to substantively different conclusions about the practical significance of a causal effect. Based on exploratory analyses of growth differences across 2019 achievement scales, further guidance is provided on when and how to use this type of aggregated benchmark for annual growth.

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Introduction

The practice of benchmarking causal effect estimates against "years of learning"–that is, expectations for grade-specific annual academic growth–is common in education. A ubiquitous first step to making sense of an estimated causal effect's practical significance is to translate the estimated causal effect into standard deviation (SD) units, often referred to as Cohen's *d* (Cohen, 1969); the causal effect in *SD* units is then referred to as an "effect size." Because the effect size is expressed in units defined by the *SD* of the outcome measure, it is no longer on the original outcome measure's scale, and effect sizes are comparable in principle across different studies using different outcome measures. However, the magnitude of effect sizes that are observed in practice can differ substantially and systematically according to subfield, research design, outcome type, population of study, and more (Cheung & Slavin, 2016; Kraft, 2020; Scammacca et al., 2015; Tanner-Smith et al., 2018; Wolf & Harbatkin, 2022). This has led to the practice

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of effect size benchmarking, in which effect sizes (in *SD* units) are compared to relevant quantities based on population-level descriptive analysis, also in *SD* units. In educational research, a peculiar challenge in effect size benchmarking is the apparent difference in the amount that students grow in their learning at different ages, which implies differences in the practical significance of causal effects at different ages. As demonstrated in a pair of highly influential papers (Bloom et al., 2008; Hill et al., 2008), according to achievement test scores, students appear to grow much more when they are in early elementary school, and much less (or not at all) by the end of high school. This is important context for making sense of any causal effect estimated in a research study; if students grow, for example, an average of 0.2 *SD* from grade 3 to grade 4, but only 0.1 *SD* from grade 5 to grade 6, then an effect size of 0.1 might naturally be interpreted as more practically significant in a study of sixth graders than in a study of fourth graders. After all, that 0.1 *SD* would appear to represent a whole year of learning for a sixth grader, compared to half a year of learning for a fourth grader. This is the basic logic of annual growth benchmarking, in which an effect of interest in *SD* units is compared to an estimate of how much students grow in a year, also in *SD* units.

In the absence of appropriate benchmarks, the effects of programs and interventions targeted to older students might be written off as insignificant, while those for programs and interventions targeted to younger students might appear more significant than they really are. What is not always appreciated is that there can be considerable systematic heterogeneity in estimates of growth even after holding student grade constant. Growth estimates may differ as a function of the population of students who took the tests (Baird & Pane, 2019) or differences in the constructs that tests have been designed to measure (Wolf & Harbatkin, 2022). Student (2022) outlines another reason why growth may differ across tests, even when they target similar constructs and populations: different psychometric designs can produce quite different estimates of annual growth, even when applied to the exact same student response data (e.g. Bolt et al., 2014; Briggs & Weeks, 2009; Li & Lissitz, 2012). This is because the creation of annual growth benchmarks relies on vertically scaled tests. Vertical scaling is a process used to place the scores of students at different grades or points in time onto a common scale, the result of which is that the scores of students in different grades are taken as measures of the student's proficiency relative to the overall targeted construct in the broad domain (e.g. math or reading) to which the scale is aligned. Vertical scaling involves numerous technical decisions about both design (e.g. what kind of test items should be used to link the vertical scale?) and modeling (e.g. which measurement model should be used for scoring?) that can affect growth patterns. The fact that different vertical scales are used to measure growth, and these differences can lead to different growth estimates, has gone relatively unexplored in the evaluation and causal inference literature in education; this study takes up this issue to provide a rationale for, and context for, updated benchmarks for growth on state achievement tests.

Study Aims

The aim of this study is to introduce a new set of empirical benchmarks for growth on state achievement tests, while recognizing the affordances and limitations of these benchmarks based on differences that may exist in growth estimates across different achievement scales. This is accomplished via three closely related sets of analyses.

The first set of analyses contributes updated estimates of annual growth on contemporary vertically scaled year-end state achievement tests in math and reading/ELA for grades 3–8. These are based on data from the 2018 to 2019 school year, the last full year of pre-pandemic schooling. In line with Bloom et al.'s assertion that benchmarks should be tailored to the outcome measure used in a study (Bloom et al., 2008, p. 319), these benchmarks are more appropriate for future studies using recent or current year-end achievement tests as an outcome measure than preexisting benchmarks, which are derived from national norms on achievement tests dating back to the 1980s and 1990s (Bloom et al., 2008).

The second set of analyses assesses differences between the benchmarks I provide and existing growth benchmarks for achievement tests, including those of Bloom et al. as well as those presented by Dadey & Briggs (2012). Dadey and Briggs' work is a meta-analysis of growth trends on vertically scaled state standardized tests in math and reading in grades 3–8 using data from the 2007 to 2008 school year. None of these tests remain in use today, and standards for math and ELA/reading have changed since 2008. I compare this study's updated benchmarks to those two studies', documenting the extent to which the size of a "year of learning" in a given subject and grade depends on which tests the benchmark is based on.

The third set of analyses explores the role that scaling plays in creating additional variability in growth estimates. I first estimate the extent to which specific design and psychometric characteristics of vertical scales are associated with differences in annual growth magnitudes, finding that the linking methods used to build a vertical scale are associated with how much students appear to grow. I then turn to estimating the overall extent to which the variability in growth estimates across different vertical scales may be attributable to differences in the scales themselves, above and beyond differences in how much students have learned relative to the targeted domain. In Dadey and Briggs' study based on data collected in 2008, each state administered its own state-specific year-end test. As a result, it was not possible to disentangle two potential sources of variability in the annual growth estimates: true differences in academic growth across states, and differences attributable to test design and scaling. With the emergence of the Smarter Balanced Assessment Consortium (SBAC) in the second half of the 2010s, it is now possible to do so, as SBAC member states all use the same year-end, vertically scaled tests targeted to the same design blueprint. Using differences in growth patterns on the National Assessment of Educational Progress (NAEP) for states that are part of SBAC and states that are not, I demonstrate the extent to which variability in growth estimates appears to be a function of differences in the design and calibration of scales, above and beyond differences in student learning. In the final section of this article, I discuss the implications of these findings for researchers attempting to make inferences about the practical significance of causal effect estimates.

Background

In the following sub-sections, I provide background on Bloom et al.'s benchmarks, how the benchmarks have been applied in recent studies, and how psychometric and design differences can affect the growth magnitudes observed on vertical scales. This background makes it clear that Bloom et al. recognized the value of using empirical benchmarks appropriate for a given outcome measure, and provides motivation for this study.

Bloom et al.'s Benchmarks

Bloom et al. (2008, p. 290) motivate the use of annual growth-based benchmarks by asking the following question: "How many standard deviations of difference represent an improvement in achievement that matters to the students, parents, teachers, administrators, or policymakers who may question the value of that intervention?" The contention that the *SD* units in which effect sizes are commonly expressed have no inherent meaning is implicit to this question. Researchers frequently rely upon Cohen (1969)'s conventions for small, medium, and large effect sizes (0.2, 0.5 and 0.8). But these benchmarks were based upon Cohen's own experience in a setting, experimental psychology, that differs in many ways from education research (Kraft, 2020). Their use as effect size benchmarks in education has the potential to label as "small" effects that are comparatively large in the distribution of effect sizes in the education research literature, and thus arguably of interest despite what would traditionally be considered a small magnitude.

To provide context for effect size interpretation that is more grounded in the field of education research and more interpretable for researchers and end users alike, Bloom et al. introduced three types of benchmarks. Two are beyond the scope of this article: benchmarks based on documented effect sizes for similar interventions, and benchmarks based on the magnitude of policy-relevant gaps in average scores between student subgroups. The third type, and the focus of this study, is benchmarks based upon average student growth on a set of nationally normed standardized tests. The benchmarks for a given subject–Bloom et al. provide benchmarks for math, reading, science and social studies–are based upon the cross-sectional standardized mean difference (d) between the test scores of students in adjacent grades. These benchmarks are meant as estimates of the causal effect of a grade-specific year of learning in its entirety–that is, there is no attempt to disentangle school-based learning from any other sources. Claimed advantages over Cohen's benchmarks include: (1) growth benchmarks are derived from empirical data in the field of education, and (2) a "year of learning" is a familiar concept for researchers, policymakers, teachers, parents, etc. that forms one of the core goals of public education. Still, Bloom et al. do not oversell the generalizability of the benchmarks, recognizing that causal effects on different outcome measures require different benchmarks: "…properly interpreting the importance of an intervention effect size requires doing so in the context of the type of outcome being measured… researchers should tailor their effect size benchmarks to the contexts they are studying" (Bloom et al., 2008, p. 307).

Perhaps the most essential finding from Bloom et al. is that according to growth norms on six nationally-normed achievement tests (seven for reading) that were particularly popular before No Child Left Behind (NCLB), younger students appear to grow much more than older students. This is shown in Figure 1, which presents Bloom et al.'s annual growth benchmarks in math and reading as bar charts. The implication for effect size interpretation is that a causal effect of a given magnitude on a standardized achievement outcome is more practically significant for older students because it represents a larger proportion of their baseline annual growth.

Applications of the Benchmarks

A defensible estimate of how much students grow is key to a number of analytic approaches in education research. In Table 1, I briefly describe and provide examples of several applications of these benchmarks, based upon a search for recent citations of Bloom et al.'s article as well as suggestions in the article itself. These include using the benchmarks in power analysis (the original context for which Cohen introduced *d* and his benchmarks), the interpretation of causal effects (in studies or meta-analyses), study selection in meta-analysis, and equating amounts of spending with amounts of student learning. In these applications, the magnitudes themselves play a substantial role in reaching a substantive conclusion about an intervention or policy. Bloom et al.'s assertion that their benchmarks are not appropriate for all outcomes implies that for a given outcome measure, there might be conceptually more appropriate benchmarks. If those benchmarks differ from Bloom et al.'s in terms of their estimated annual growth magnitudes, then the substantive research findings described in Table 1 change accordingly. Thus, benchmarks for growth on state year-end tests fill a real need in the education research literature for studies that use scores on these tests as an outcome measure.

How Vertical Scaling Decisions Can Create Differences in Growth on Similar Measures

Above, I noted that Bloom et al. encouraged researchers to tailor their growth-based effect size benchmarks to their outcome measures whenever possible. One reason why this is so important is that growth on the pre-NCLB tests whose norms are the basis for Bloom et al.'s benchmarks may differ from growth on more contemporary measures, such as state achievement tests. This may be a result of differences in student learning on the content covered by these tests and/or technical scaling differences between the

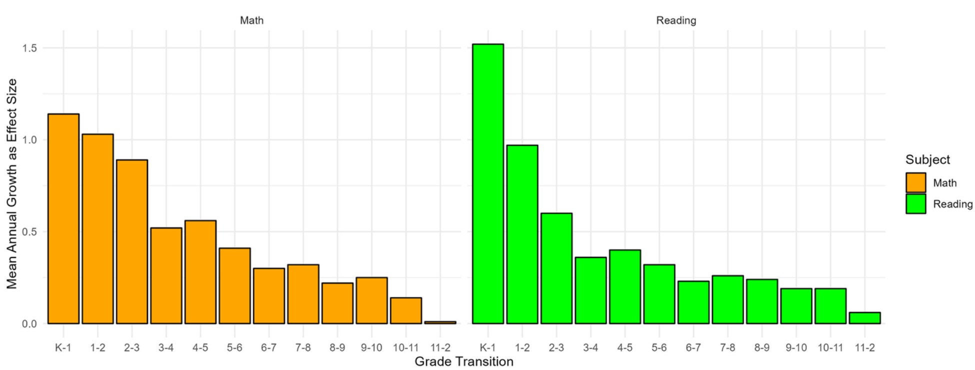

Figure 1. Bar chart of Bloom et al.'s empirical benchmarks for growth using Cohen's *d*.

Table 1. Uses of Bloom et al.'s growth benchmarks.

Application	Example
To inform the minimum detectable effect for power analysis	Described in Bloom et al. (2008, p. 290)
Assessing the practical significance of an estimated causal effect	"The decline of 0.3 standard deviations in math is roughly equivalent to one-half of a year's worth of learning in fifth grade and approximately a full year's worth of learning in middle and high school. A decline of 0.1 to 0.2 standard deviations in ELA translates to between one-third and one-half of a year's worth of learning, depending on the grade level." (Kogan & Lavertu, 2022, p.10)
Meta-analysis study selection	"Since younger students in lower grades (i.e., K–3) are expected to see more considerable yearly gains in math than their older peers (Bloom et al., 2008)… their reported [effect size] may not adequately reflect the impact on older students… we address the potential bias by only evaluating studies addressing students with [mathematics difficulties] in Grades 4 to 12." (Myers et al., 2023, p. 7)
Fiscal analyses related to student learning	"…The authors[a] find that an additional $1,000 of annual per pupil spending yields .012 *SD* of achievement growth for low-income students and .007 *SD* for non-low-income students… average per pupil expenditures in the United States is about $13,000 per pupil. On average, in a year, student achievement increases by about .6 *SD* (Bloom et al., 2008). Over the course of the year, then, those $13,000 of educational expenditures can be thought to yield approximately .15 *SD* of achievement growth… with the remaining .45 coming from family and other nonschool inputs." (Shores & Steinberg, 2022)

Note: [a]Is in reference to Jackson and Mackevicius (2021).

tests. For the purposes of their analysis, Bloom et al. state the assumption that the latter is *not* an issue and do not investigate further:

> Scaled scores for these tests were created using Item Response Theory methods. Ideally, these measure "real" intervals of achievement at different ages so that changes across grades do not also reflect differences in scaling. Investigation of this issue is beyond the scope of this article. (Bloom et al., 2008, p. 297)

This idealization is not borne out by recent research on vertical scaling. While a complete review of this literature is beyond the scope of this article (see Reckase, 2010 for a review of many of the psychometric challenges of vertical scaling), in this section, I review two psychometric decisions that have been shown to affect growth patterns on vertically scaled tests: common item linking design and choice of measurement model. I focus on these because, as shown below, the achievement scales used in this study differ in these two dimensions. Many other aspects of psychometric modeling can also affect growth magnitudes, but could not be included in this study. For example, scoring method (i.e. maximum likelihood vs. methods using a Bayesian prior) can affect how much students appear to grow (Briggs & Weeks, 2009), but among the scales reviewed for this study, all used the same scoring method, maximum likelihood.

Some familiarity with Item Response Theory and vertical scaling is assumed. For a review of vertical scaling and effect size benchmarks targeting an audience with no psychometric background, see Student (2022). More in-depth reviews of vertical scaling methods are available in Briggs & Weeks (2009), Tong and Kolen (2007), and chapter 9 of Kolen and Brennan (2014). The overarching point of this review of psychometric decisions that go into vertical scaling is this: tests frequently differ in the psychometric designs and methods used to construct and score them, and these decisions can have direct implications for the growth magnitudes on which benchmarks like Bloom et al.'s are built.

Vertical Scale Linking Design

Vertical scales almost always rely upon a common item, nonequivalent group design to link the scores of students in different grades (Kolen & Brennan, 2014). This refers to a design in which students in adjacent grades (nonequivalent groups) take a set of the same items (common items), and their differential performance on those items is used to make mathematical adjustments that result in scores being on a common scale. These items are administered as a subset of the full test. Because these items are used to estimate growth, *which* items are used as common items is crucially important to vertical scale construction.

Tests can and do differ in the grade level of the common items used for scale construction. For example, consider linking scales for grades 3 and 4 to create a single (vertical) scale for scores on both a grade 3 and grade 4 test of mathematics. One can use common items aligned to grade 3 standards ("backward" design; Briggs & Dadey, 2015), common items aligned to the grade 4 standards ("forward" design), or a combination of grade 3 and grade 4 items ("mixed" design) as common items. Each of these designs reflects a different conception of what "growth" is, and scales that differ in common item grade alignment may produce different growth magnitudes, as noted by Kolen & Brennan (2014). Specifically, a forward design would be expected to produce the largest estimates of growth, while a backward design is expected to produce the smallest. This is akin to an item-level heterogeneous treatment effect (Gilbert et al., 2023) in which a year of classroom learning has more impact upper-grade items than on lower-grade items. This reflects a larger ongoing concern in vertical scaling about "construct shift," the extent to which differences in the content of the domain being measured (e.g. mathematics) across grades can influence the extent to which students appear to grow (Hansen & Monroe, 2018; Li & Lissitz, 2012; Strachan et al., 2021). However, construct shift is difficult to assess (to the extent that psychometricians even agree on a definition of what it is), while the choice of common item linking design is easily assessed from technical manuals and has clear empirical implications. This study therefore attends only to common item linking design as a point of differentiation across scales.

Differences in Item Response Theory Calibration Approaches

Putting aside issues of common item linking design, the psychometric methods used to construct a vertical scale can also affect growth magnitudes. A major differentiating point is the measurement model used to estimate item parameters and generate scale scores. The two most popular measurement models for vertical scaling, as found in this study, are the Rasch model (Rasch, 1960).[1] and the three-parameter logistic Item

[1]Although the details are beyond the scope of this article, the Rasch model is particularly suitable to vertical scaling applications because when item response data fit the model, parameters produced by the Rasch model have the equal-interval (Stevens, 1946) metric properties assumed by most parametric statistics (Ballou, 2009; Briggs, 2013).

Response Theory model (3PLM) (Birnbaum, 1968). Both the Rasch model and the 3PLM are appropriate for dichotomously (i.e. correct-incorrect) scored items. Where needed, they are used alongside a model for polytomous (i.e. partial-credit) items: the Partial Credit Model (PCM; Masters, 1982) extends the Rasch model, while the Generalized Partial Credit Model (GPCM; Muraki, 1992) is typically used with the 3PLM. As demonstrated in Briggs & Weeks (2009), the choice whether to use the Rasch model/PCM or 3PLM/GPCM can affect the amount of growth one finds on vertical scales. They found differences of up to 0.2 *SD* when comparing growth according to the Rasch model/PCM versus the 3PLM/GPCM when applied to the same underlying item response data from a state test of reading for grades 3–6. These differences represent a significant portion of a year of learning as calculated by Bloom et al. (2008) in grade 3–6 reading, where benchmarks for annual growth range from 0.32–0.40 *SD*. Finally, a third measurement model, the two-parameter logistic model (2PLM), is also used in several prominent testing programs such as SBAC and NAEP, in concert with a model such as the GPCM for items with more than two scoring categories.

Scale Evaluation Challenges

These design and implementation differences exist because there is no consensus on how to approach the aspects of vertical scale construction outlined above or to evaluate the overall quality of a vertical scale (see, e.g., Briggs, 2013, contrasted with Yen, 1986). One can make assumptions about growth, such as students in higher grades scoring higher than students in lower grades. These assumptions are not always borne out in practice, though, as Dadey and Briggs (2012) document instances of state-level negative growth in reading from grades 4–5 and 5–6 and Briggs and Dadey (2015) show that "reversals" in which common items appear harder for upper-grade students often have a coherent, content-based explanation. It is possible to apply non-linear transformations during scale construction to ensure that a vertical scale meets a priori assumptions about patterns in student learning (Kolen & Brennan, 2014), but doing so is not in itself verification that growth actually follows the hypothesized trajectory across years (per Dadey and Briggs).

Methods

This section outlines the methods used in each set of analyses. All of the analyses in this study are based upon a set of estimates of annual growth in standard deviation units on each 2019 vertical scale for which I was able to acquire data. Before describing the analyses below, I first describe how I went about constructing this dataset, and make a key distinction between estimates at the *state* level and estimates at the *scale* level.

I began by identifying all states that used a vertical scale for their grade 3–8 math and ELA tests in the 2018– 2019 school year using publicly available documentation including technical manuals, score interpretation guides and cut score documents. For each state, I then gathered basic descriptive statistics for the scale scores of all students in the state for grades 3–8 in math and reading/ELA: means, standard deviations and sample sizes by grade level. This information was gathered from technical manuals, state data portals and public data requests submitted to state departments of education. I was ultimately able to gather such data for 20 of 25 states identified as using vertical scales in 2019. For each state for which I was able to collect score data, I calculated cross-sectional growth in *SD* units for each pair of adjacent grades using standard deviations pooled across adjacent grades. That is, for a state *i*, growth in subject *s* from grade *g*–1 to grade *g*, Y_{sig}, is calculated as:

$$Y_{sig} = \frac{\bar{\theta}_{sig} - \bar{\theta}_{si(g-1)}}{\sqrt{\frac{\sigma^2_{sig} + \sigma^2_{si(g-1)}}{2}}} \quad (1)$$

Where $\bar{\theta}_{sig}$ is the mean scale score in subject *s*, state *i* and grade *g*, and σ^2_{sig} is the variance of those scores. This is the same approach to calculating growth in *SD* units used in Dadey and Briggs (2012), as suggested by Yen (1986).

The unit of analysis in most of the analyses in this article, however, is *scales*, not states. When Dadey and Briggs (2012) conducted their study, every state used its own unique vertical scale, so there was a one-to-one relationship between states and scales. In 2019, 10 of the 20 states for which I was able to collect data were members of SBAC, meaning that they all used the same scale for their math and reading achievement tests. Before creating updated growth benchmarks, then, a necessary intermediary step was to calculate estimated annual growth on SBAC, as a function of estimated annual growth in each SBAC state.

To create a single estimate of growth on SBAC in each grade/subject, I took the mean of growth in state-specific *SD* units in each SBAC state within that grade/subject, weighted by the state-specific sample size. Although national summary statistics for scale scores on SBAC summative tests are available in the SBAC technical report (Smarter Balanced Assessment Consortium [SBAC], 2020), I opted to create a weighted average from the raw data because I was unable to acquire scale score information from every SBAC state, and using this weighted average made it possible to use the exact same analytic sample across all analyses described below, whether the unit of analysis is scales or states. In the analyses described below, the unit of analysis is scales in every set of analyses except the last, which compares state-level estimates of growth between the state test and NAEP.

Analysis 1: Benchmarks for Annual Growth on State Achievement Tests in Math and ELA

I first present descriptive information about the mean, standard deviation, minimum and maximum of growth in each subject and grade pair Y_{sig} across all vertical scales used on state tests in 2019. The means are intended as a new set of empirical benchmarks for growth that will be useful for researchers using state achievement tests as outcome measures. The standard deviations are included to emphasize the considerable variability of growth around each mean. Appendix C of the online supplementary materials discusses the choice to report these descriptive statistics as benchmarks, which is in line with Dadey and Briggs (2012) but contrasts with Bloom et al. (2008)'s use of a meta-analytic random effects estimator. Subsequent analyses serve to contextualize these benchmarks.

Analysis 2: Comparison with Prior Benchmarks

These analyses entail comparing my findings to prior studies answering similar questions with different tests (Bloom et al., 2008; Dadey & Briggs, 2012). First, I compare benchmarks for annual growth in *SD* units in each grade/subject combination using 2019 state test scales (this study), 2008 state test scales (Dadey & Briggs) and pre-NCLB national norms (Bloom et al.). I note where estimates of growth differ across my benchmarks, Dadey and Briggs's, and Bloom et al.'s, as well as where growth appears most similar across the three.

I also compare the extent of deceleration implied by my own benchmarks from Aim 1 to the deceleration documented by Bloom et al. and Dadey and Briggs. It is important to understand if deceleration on contemporary tests looks similar to the deceleration that partially motivated Bloom et al.'s original work; after all, deceleration is the main reason why Bloom et al. argued that growth-based benchmarks hold practical value. I do this by fitting linear regressions where the data points are the grade-specific benchmark values for one study in one subject, then comparing the results across studies. Analyses are based on the following simple linear regression model, which was fit six separate times (three studies by two subjects):

$$\overline{Y_g} = \beta_0 + \beta_1 grade_g + \epsilon_g \tag{2}$$

Where $\overline{Y_g}$ is average growth from grade g–1 to g in *SD* units (that is, the benchmark), β_0 is the intercept, $grade_g$ is the grade level minus three (so that $grade_g = 0$ for grade three, 1 for grade four, etc.), β_1 is the coefficient representing the linear association between one additional grade level and predicted growth in *SD* units, and ϵ is an error term assumed to have constant variance with an expected value of zero. β_1 is the coefficient of interest for comparing deceleration trends from this study against others; a negative β_1 coefficient indicates deceleration, while a positive β_1 would represent acceleration. Figure 2 shows Bloom et al.'s benchmarks as scatterplots; the blue regression lines superimposed on these plots are β_1.

Comparing β_1 across the benchmarks from this study, Dadey and Briggs's study, and Bloom et al.'s study answers the question: compared to prior benchmarks, does growth appear to decelerate more or less using benchmarks based on 2019 test scores?

Because variability in growth is of interest for the purpose of this study, I then fit this same model individually for math and reading on each 2019 vertical scale separately, using the model:

$$Y_g = \beta_0 + \beta_1 grade_g + \epsilon_g \tag{3}$$

The only difference between (2) and (3) is the notation used for the outcome: Y_g vs. $\overline{Y_g}$. Y_g is used here to clarify that this model is being fit with data from one scale at a time; $\overline{Y_g}$ was used in (2) to indicate that the outcome is the benchmark itself, which is an average across scales. This model was thus fit 22 times: 11 scales by two subjects.

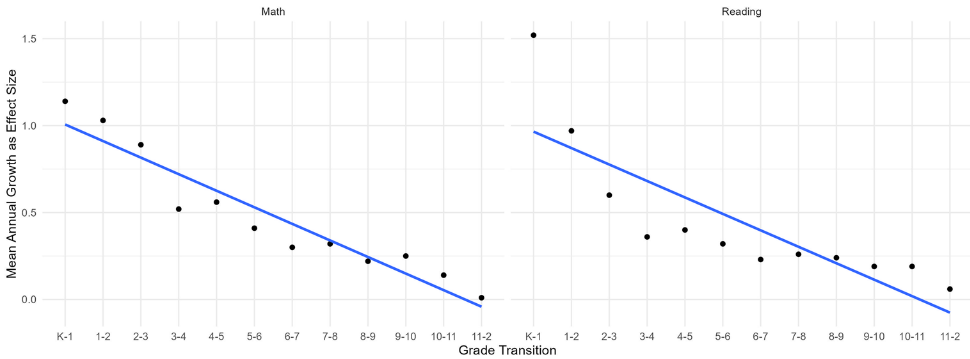

Figure 2. Linear representation of deceleration in Bloom et al. (2008)'s benchmarks for growth.

The distribution of resulting β_1 values in each subject shows not only what deceleration looked like in 2019 in general, but also how variable it was. Descriptive statistics summarize this distribution. I use the method of separate within-scale regressions for comparability to Dadey and Briggs (2012, p. 12), who summarized coefficients from separate regressions as a representation of variability in the growth trajectories.

Additionally, Appendix D in the online supplementary materials contains two further analyses comparing growth on 2019 scales to the 2008 scales used by Dadey and Briggs (2012). These concern the phenomena of variability in growth estimates for each grade pair and scale expansion/shrinkage.

Analysis 3: Exploring the Role of Scaling

While the first two sets of analyses concern the creation of new benchmarks for growth on state tests of math and reading, and comparison of these benchmarks to prior benchmarks in the same subjects, these analyses investigate *why* estimates of growth vary across scales. To approach this question, I use two exploratory approaches intended to address specific aspects of the relationship between scale choice and annual growth estimates.

Regressing Growth on Scale Characteristics

This analysis explores the extent to which specific characteristics of vertical scales are associated with growth magnitudes in *SD* units. In the Background section above, I reviewed two key aspects of vertical scaling that affect growth magnitudes: measurement model (Rasch/2PLM/3PLM) and linking approach (forward, backward, mixed or nonlinear)[2]. Although these are far from the only psychometric distinctions that affect

[2]In all cases, scales using the Rasch model used the PCM, and in all cases, scales using the 3PLM used the GPCM. Scales using the 2PLM used either the GPCM or the Graded Response Model (Samejima, 1969)–they were coded with the same "2PLM" variable for this analysis as only two scales used the 2PLM. All scales I reviewed used mixed or forward linking designs, with one exception–one scale used a design where non-linear transformations ensure that growth on that scale follows a pre-specified grade-to-grade pattern, a method also used on some of the scales analyzed by Dadey and Briggs.

growth estimates, technical manuals for the scales in this study contain information about these aspects of vertical scale design and construction, and both would be expected to affect growth estimates.

To assess the association between linking direction/measurement model and growth on vertical scales, I again use linear regression. Here, separately by subject, I fit four models that predict growth in *SD* units. As in the analyses above, the unit of analysis here is scales, and data from all scales in one subject are used to fit the models.

The first model predicts growth in *SD* units as a function of grade:

$$Y_{ig} = \beta_0 + \beta_1 grade_g + \epsilon_{ig} \qquad (4;\ \text{Model 1})$$

This model provides a baseline against which to compare subsequent models. Here, R^2 is the primary result of interest, as this represents the extent to which growth in *SD* units on scale i in grade g, Y_{ig}, can be predicted purely as a function of the grade level of the test within the given subject. Adding in variables then tells us how much more variance those variables explain, above and beyond grade level.

The second model adds linking design as a predictor.

$$Y_{ig} = \beta_0 + \beta_1 grade_g + \alpha_1 forward_i + \alpha_2 nonlinear_i + \epsilon_{ig} \qquad (5;\ \text{Model 2})$$

The third model adds measurement model as a predictor.

$$Y_{ig} = \beta_0 + \beta_1 grade_g + \alpha_3 2PLM_i + \alpha_4 3PLM_i + \epsilon_{ig} \qquad (6;\ \text{Model 3})$$

Finally, the fourth model includes both measurement model and linking design as predictors.

$$Y_{ig} = \beta_0 + \beta_1 grade_g + \alpha_1 forward_i + \alpha_2 nonlinear_i + \alpha_3 2PLM_i + \alpha_4 3PLM_i + \epsilon_{ig}$$

(7; Model 4)

The reference linking design is thus mixed, with α_1 and α_2 representing the conditional association between growth and the use of forward linking or non-linear scale construction, respectively. Similarly, the reference measurement model is the Rasch model, with α_3 and α_4 representing the conditional associations between the use of the 2PLM and 3PLM models and growth, respectively.

In these analyses, regression is being used strictly to describe the conditional relationships between the predictor variables and growth in *SD* units on different scales. I do report statistical significance based on cluster-robust CR2 standard errors to contextualize the findings in terms of likelihood of the findings under a hypothesis of no true association, but I also emphasize the extent to which the addition of predictors leads to an increase in model R^2, and the direction and magnitude of the coefficients.

Comparisons to Growth on NAEP

The second exploratory analysis is, as noted above, the only analysis where the unit of analysis is *states*, rather than scales. This analysis makes use of information about whether states used the SBAC summative math and reading tests because states using SBAC summative tests all employ the same underlying scale.[3] This analysis represents an attempt to understand the extent to which the differences in growth across scales documented in this study are a direct consequence of the use of different scales, above and beyond underlying differences in learning across states. I investigated this because the variability in growth introduced by the use of different scales underscores the importance of the standard deviations reported in Analysis 1. Using a single point estimate for annual growth without acknowledging the scale-to-scale variability around that estimate can lead to overstating the precision of a "year of learning" effect size translation, an issue to which I return in the Findings and Discussion sections.

For these analyses, I make comparisons between 2019 state test scores and results on the 2019 NAEP math and reading tests in the same states. The vertically scaled portion of NAEP includes tests for students in grades 4 and 8. That is, the only cross-sectional growth that can be estimated from NAEP results is growth from grade 4 to grade 8, with no breakdown of where that growth occurs in the grades between. For this analysis, I first calculate cross-sectional growth for every state, SBAC or not, included in this study from grade 4 directly to grade 8. That is, using the notation above, I calculate growth from grade 4 to 8 as

$$Y_{sj[4,8]} = \frac{\bar{\theta}_{sj8} - \bar{\theta}_{sj4}}{\sqrt{\frac{\sigma^2_{sj8} + \sigma^2_{sj4}}{2}}} \tag{8}$$

This is calculated for both the state j achievement test in subject s and NAEP in the same subject and state. Because NAEP targets the same subjects as state tests, NAEP growth is expected to explain at least some of the variance in growth on state-level tests. To assess this, I present the Pearson correlation r between grade 4 and 8 growth on state tests and on NAEP along with scatter plots disaggregated by SBAC membership and subject. These correlations can be squared to estimate the variance explained by NAEP growth in SBAC vs. non-SBAC states in math or reading. The difference in these R^2 values is representative of the extent to which the use of different scales introduces additional variance into state-level growth estimates; for SBAC states, all of the variance in growth reflects differences in learning, so the amount of this variance explained by NAEP serves as a baseline. Against this baseline we compare the amount of variance explained by NAEP among states that use different scales, where variance in growth reflects both learning differences and the use of different scales. If R^2 is lower when regressing growth in states using their own tests

[3]There is one exception here, Indiana, who used the underlying SBAC vertical scale but applied different linear transformations to create their final score scale for reporting. Because the transformation is linear, this distinction disappears once growth is translated into SD units.

Table 2. Benchmarks for growth on 2019 state achievement tests in Math and ELA and associated descriptive statistics.

Subject	Grade pair	Mean	*SD*	Min	Median	Max	*N*
ELA	3–4	0.51	0.11	0.38	0.49	0.77	11
ELA	4–5	0.39	0.14	0.17	0.38	0.63	11
ELA	5–6	0.24	0.15	−0.03	0.23	0.55	11
ELA	6–7	0.28	0.20	0.04	0.25	0.56	11
ELA	7–8	0.23	0.10	0.02	0.24	0.41	11
Math	3–4	0.58	0.19	0.22	0.63	0.82	11
Math	4–5	0.50	0.15	0.27	0.47	0.71	11
Math	5–6	0.31	0.24	−0.03	0.23	0.67	11
Math	6–7	0.29	0.19	−0.01	0.24	0.51	11
Math	7–8	0.41	0.15	0.11	0.46	0.63	11

on those states' NAEP scores than it is when regressing growth in SBAC states on their NAEP scores, then that decrease in R^2 represents variability potentially attributable to differences in scale design and calibration, as well as other differences between state tests.[4] If the difference in R^2 is substantial, then this is evidence that it is important to account for the variability of growth across scales, not just a single point estimate of mean growth, in "years of learning" translations of effect sizes.

Findings

Analysis 1: Benchmarks for Annual Growth on State Achievement Tests in Math and ELA

Table 2 presents descriptive statistics for growth in *SD* units on 2019 state achievement test scales. As noted above, the means in this table are suggested as new benchmarks for annual growth against which effect size estimates might be compared, and this overall average trajectory is plotted on top of each individual scale's trajectory in Figure 3.

These means imply general growth deceleration in both subjects across these grades; as detailed below, this is roughly in line with past benchmarks, though meaningful differences do exist. In particular, note growth in grade 7–8 math, which represents an acceleration compared to grade 6–7. There is considerable variability around these means, as well. This is perhaps made clearest by considering the proportion of a "year of learning" represented by each *SD*: *SD*s range from about 20% to about 80% of a year of learning. One scale's "year of learning" is often much larger than another's,

[4]It is important to recognize here that NAEP and state tests do not target identical constructs; NAEP targets the NAEP subject domain content frameworks, while state tests target state-specific content standards. Given this, differences in standards' alignment with NAEP frameworks may also contribute to differential R^2 when looking at the predictiveness of NAEP results for growth in SBAC vs. non-SBAC states. Additionally, concerns have been raised about treating differences between grade 4 and 8 scores on the NAEP scale as growth magnitudes (Haertel, 1991; Thissen, 2012). This should not be an issue for these analyses, as I do not make claims about whether students grow more or less on NAEP than on state tests–only whether growth on NAEP is more variable than on state tests. Finally, different tests may be differentially vulnerable to score inflation (Koretz, 2005), though I am not aware of any study of this phenomenon for the post-NCLB scales used in this study. Given these threats to validity, the analyses in this paper represent an upper limit on how much variability in growth can be attributed to the use of different scales in different states.

and Analysis 3 below gives us reason to believe that this is not a matter of random chance.

Analysis 2: Comparison with Prior Benchmarks

Comparing Means

Figure 4 presents annual growth estimates in ELA/reading and math from this study, Dadey and Briggs (2012), and Bloom et al. (2008). This comparison shows that although overall patterns across all three sets of benchmarks are roughly similar, with decelerating growth from grades 3–8, benchmarks constructed from different scales do lead to different annual growth estimates. In ELA/reading, perhaps the most striking finding is that using state achievement tests, whether 2019 or 2008, leads to an estimate of grade 3–4 growth that is much larger than Bloom et al.'s benchmark of 0.36. Yet, the 2019 benchmarks contain the smallest estimate of grade 7–8 growth in ELA, creating the sharpest contrast between early and later grades of the three sets of benchmarks. Turning to math, differences also emerge. Here, 2019 state achievement tests produce the largest estimates of growth in both grades 3–4 and grades 7–8. Overall patterns across all grades appear roughly similar across the three sets of benchmarks, though grade 5–6 growth on 2019 state tests is also markedly lower than on either prior set of benchmarks.

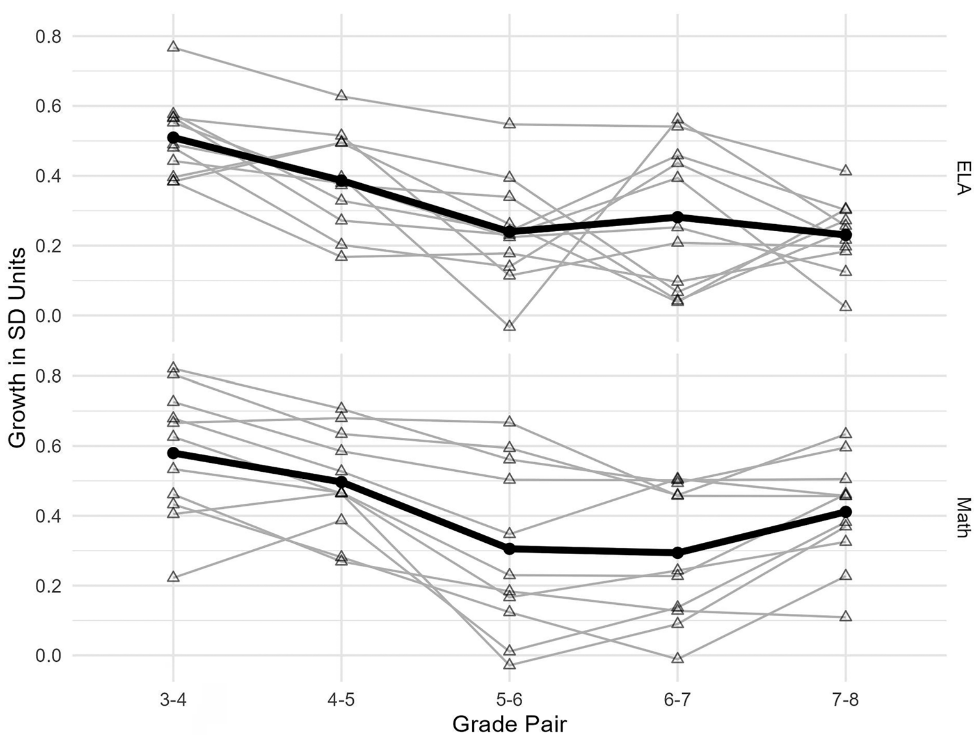

Figure 3. Scale-specific and average trajectories of growth on 2019 state achievement tests.

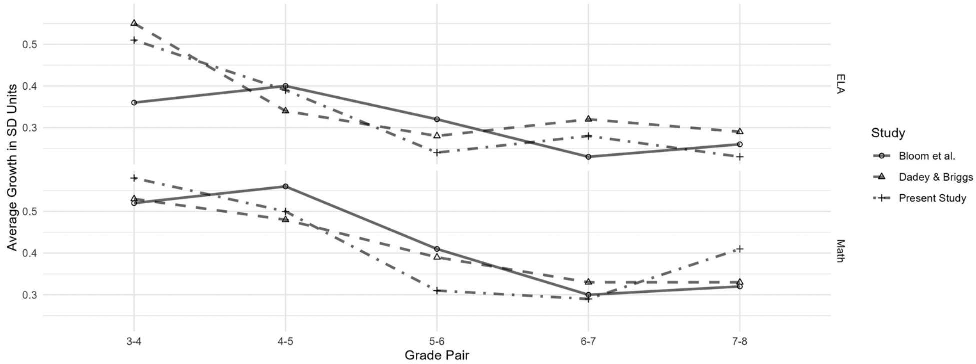

Figure 4. Comparison of benchmarks for growth in *SD* units by grade and subject.

Table 3. Comparison of linear growth trends across benchmarks from three studies.

	ELA/reading		Math	
Study	β_0	β_1	β_0	β_1
Present study	0.464	−0.067	0.528	−0.055
Dadey & Briggs	0.464	−0.054	0.522	−0.055
Bloom et al.	0.388	−0.037	0.554	−0.066

Note: β_0 is the intercept (predicted grade 3–4 growth), while β_1 is the slope for changes in growth across grades, with negative values representing growth deceleration.

Patterns of Deceleration

Table 3 below presents a comparison of the model coefficients from a simple regression of growth on grade level across each of the three studies (Equation 2 above). Estimates of β_1, the slope, summarize patterns of deceleration across all three studies. In both subjects, deceleration (a negative β_1 coefficient) is present in every model. The original motivation for Bloom et al.'s study was at least partly the fact that younger students appear to grow more on achievement tests than older students do. This remained the case on 2008 state achievement tests, as well as on 2019 state achievement tests. Deceleration in ELA/reading appears steepest in the benchmarks I presented above, and least steep in Bloom et al.'s original benchmarks. In contrast, deceleration in math is identical between Dadey and Briggs's study and the benchmarks I presented above, and is less steep than the pattern in Bloom et al.

I also fit a similar model for each individual scale used in this study (Equation 3) to get a sense of the extent to which patterns on individual scales vary. Table 4 presents descriptive statistics for the intercept and the slope (representing deceleration) across all 11 models in each subject (one per scale), while Figure 5 shows all 11 linear trajectories on a single plot along with the overall trajectory from Table 3. This table shows that in ELA, deceleration is present on every scale used in the study, though the extent of deceleration does differ among scales, as shown in Figure 5. In math, deceleration is nearly ubiquitous, although one state does not display deceleration, with a very small increase in growth from grades 3–4 to grades 7–8. Visually and according to Table 3, more variability is present in the intercepts. That is, much of the variability in overall growth trends across scales appears to be a result of the fact that growth in grades 3–4 differs substantially across scales. From here, growth on

Table 4. Descriptive statistics for coefficients of regression of growth on grade level based upon growth on 11 scales in 2019.

	ELA		Math	
Statistic	β_0	β_1	β_0	β_1
Mean	0.462	−0.066	0.525	−0.054
SD	0.113	0.027	0.181	0.026
Min	0.296	−0.104	0.214	−0.089
Max	0.738	−0.029	0.768	0.007
N	11	11	11	11

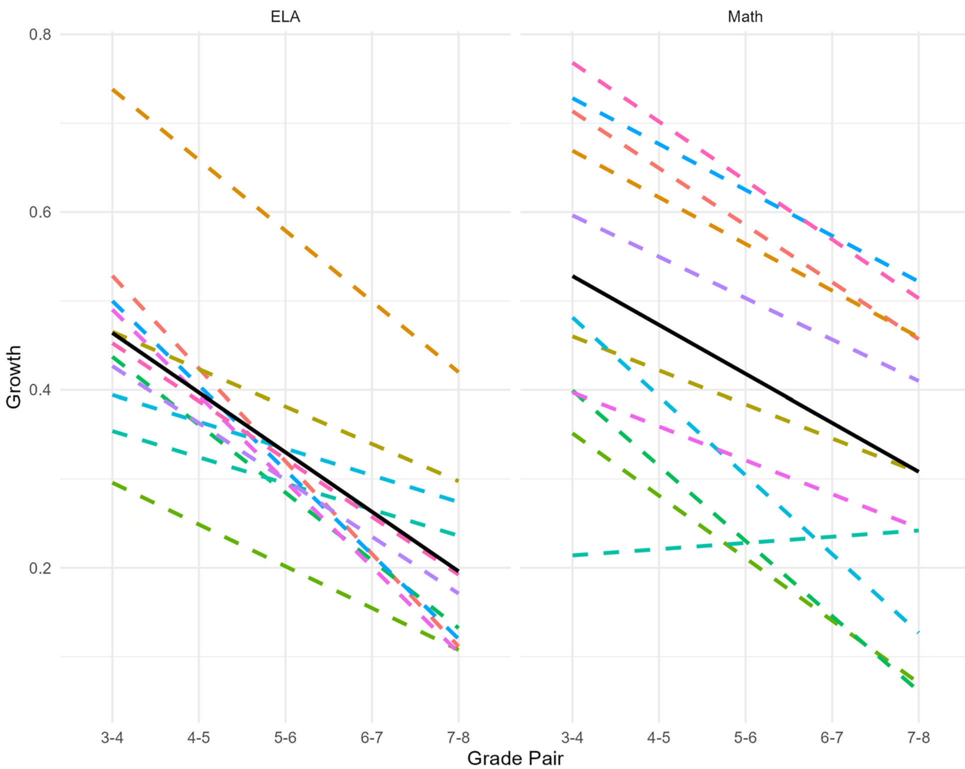

Figure 5. Scale-specific and overall linear growth trajectories.

most scales tends to follow a roughly similar pattern, but different starting points (and departures from the linear trend) drive meaningful differences in growth magnitudes across scales. There does appear to be less variability in the extent of deceleration (i.e. *SD* of β_1) in the present study than Dadey and Briggs (2012) found in their analysis: Dadey and Briggs report an *SD* of 0.04 for reading, while the present *SD* for ELA is 0.027; for math, Dadey and Briggs report 0.05, while I found an *SD* of 0.026.

Analysis 3: Exploring the Role of Scaling

Regression of Growth on Scale Characteristics

Recall that the four models for this analysis built up from a model that predicts growth from grade alone to one predicting growth from grade and two technical scale characteristics: measurement model (Rasch, 2PLM or 3PLM) and scale construction approach (forward linking, mixed linking or non-linear). Table 5 presents the results of fitting these regression models for ELA, while Table 6 presents the same for math. Note that I was unable to find the required information for one scale, so all analyses are based on 10 scales, not the full 11 used elsewhere. With ten scales and five grade pairs per scale (3–4, 4–5, 5–6, 6–7, 7–8), each model was fit to 50 observations.

Across all four models for ELA, the (conditional) association between grade and growth remains identical: growth in *SD* units is predicted to decrease by 0.070 *SD* units for each grade pair above 3–4. Looking to the R^2 for model 1, grade alone explains about 31% of the total variance in growth across scales. Model 2 adds in linking design as a predictor of growth. This addition results in an R^2 of 0.534, a marked increase. As indicated by the magnitude and significance of the α_2 coefficient, this appears to be driven in large part by modeling the use of a non-linear (a priori) vertical scale design by one state. This design involves pre-specifying the distribution of scores in each grade so that they resemble growth on prior scales, and as such, is not a "linking" design at all, but an alternative to linking that pre-specifies growth magnitudes irrespective of any actual student item responses (as noted by Dadey &

Table 5. Regression analysis predicting growth from scale characteristics, ELA.

		Conditional association with ELA growth in *SD* units			
Coefficient	Predictor	Model 1	Model 2	Model 3	Model 4
β_1	Grade	−0.070**	−0.070**	−0.070**	−0.070**
α_1	FL		0.047		0.040
α_2	NLT		0.300**		0.294**
α_3	2PLM			0.147	0.013
α_4	3PLM			0.037	0.018
β_0	Intercept	0.473**	0.419**	0.425**	0.412**
# Obs.		50	50	50	50
r^2		0.307	0.534	0.392	0.536
Adj. r^2		0.293	0.504	0.352	0.483
RMSE		0.149	0.122	0.139	0.122

Note. FL: forward linking; NLT: Non-linear transformation; RMSE: root mean squared error; $^{**}p < 0.01$.

Table 6. Regression analysis predicting growth from scale characteristics, Math.

		Conditional association with ELA growth in *SD* units			
Coefficient	Predictor	Model 1	Model 2	Model 3	Model 4
β_1	Grade	−0.060**	−0.060**	−0.060**	−0.060**
α_1	FL		0.280**		0.267′
α_2	NLT		0.298**		0.334**
α_3	2PLM			0.030	−0.048
α_4	3PLM			0.127	0.002
β_0	Intercept	0.556**	0.387**	0.487*	0.398**
# Obs.		50	50	50	50
r^2		0.170	0.622	0.250	0.626
Adj. r^2		0.152	0.597	0.201	0.583
RMSE		0.188	0.127	0.179	0.126

Note. FL: forward linking; NLT: Non-linear transformation; RMSE: root mean squared error; $'p < 0.1$; $^{*}p < 0.05$; $^{**}p < 0.01$.

Briggs [2012], this practice was used for two of the scales in their analysis as well). Growth in this state in ELA is large compared to growth on all other ELA scales, shown in Figure 6; Figure 6 does also show that growth generally appears greater in ELA on scales using forward linking compared to mixed, but by a smaller magnitude that is not statistically significant. Model 3 uses measurement model as a predictor, and although both α_3 and α_4 are positive, neither is significant and the jump in R^2 is not as dramatic as that for Model 2. Finally, Model 4 shows that once linking design is accounted for, measurement model appears to explain essentially no additional variance in growth, with an adjusted R^2 (which penalizes additional predictors) below that of Model 2.

Turning to math in Table 6, linking design again appears to be much more strongly associated with growth magnitudes than measurement model. Figure 6 shows that unlike in ELA, this is attributable in large part to the magnitude of growth on forward-linked scales, which is reflected in the magnitude of α_1. This is highlighted further by the fact that grade alone–Model 1–explains just 17% of the variance in growth estimates, while Model 2's R^2 is 0.622–a nearly fourfold increase. Meanwhile, the R^2 for Model 3 is only 0.250, and Model 4 indicates that measurement model again tells us essentially nothing above and beyond linking design. The statistical significance of α_1 in Model 4 is borderline, but it is significant at the $p < 0.01$ level in Model 2, where the non-significant predictors related to measurement model choice are omitted.

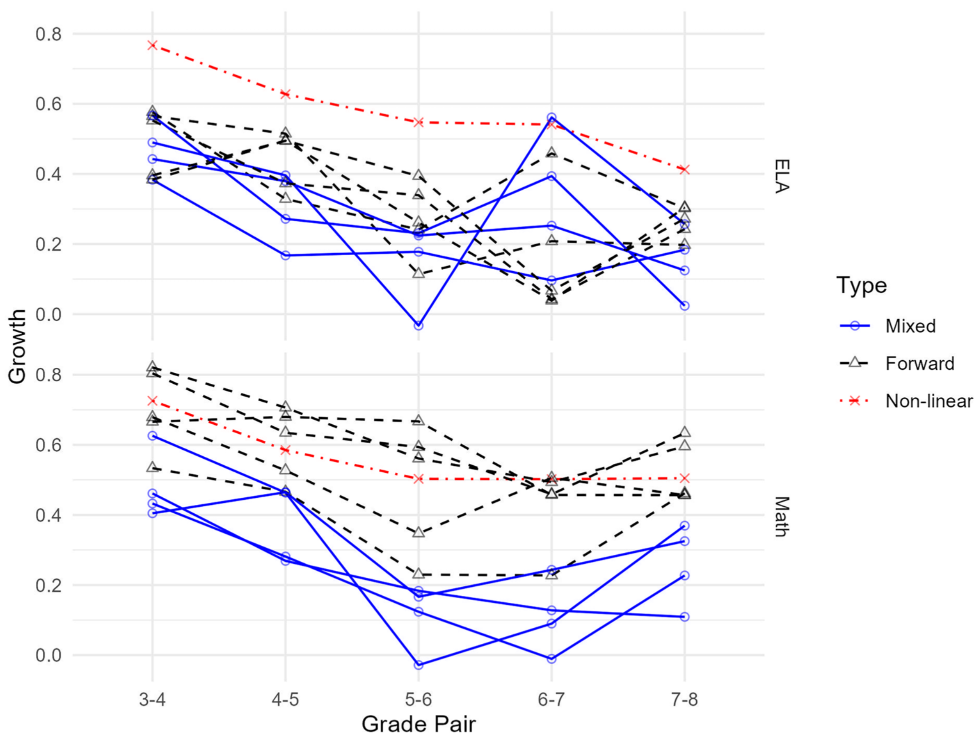

Figure 6. Growth on state achievement scales by scale design.

Comparison with Growth on NAEP

Figure 7 shows the linear association between grade 4–8 growth in *SD* units on state tests and grade 4–8 growth in *SD* units on NAEP, disaggregated by subject and whether or not each state used the SBAC scale in 2019 (among the states for which I was able to gather data, ten used SBAC and ten did not, so there are ten points per panel). As a reminder, "growth in *SD* units" here is the mean difference in scores between grade 4 and grade 8 students, divided by the pooled *SD* of scores in those two grades only (see Equation 8). This figure demonstrates the extent to which the use of different scales appears to increase variability in growth estimates compared to a set of states that all use the same scale.

In ELA/NAEP reading, among states that used the SBAC scale, the correlation of growth on the state test with growth on NAEP is 0.48 (R^2 = 0.23). Meanwhile, for non-SBAC states, the correlation is *negative*: −0.23 (R^2 = 0.05). In math, the correlation for SBAC states is 0.81 (R^2 = 0.66), compared to 0.25 (R^2 = 0.06) for non-SBAC states. This leads to two main findings. First, these findings are in line with the assertion that when states use different scales, the resulting growth estimates are much more variable than they would be if growth were measured on the same scale. Figure 8 shows growth on state tests across grades in 2019 by state, using color and line type to differentiate between SBAC states and states using their own scales (here, again, there are 20 total states with 10 using SBAC). This shows that the findings using NAEP data are not peculiar to grades 4–8; in every grade, growth estimates on SBAC

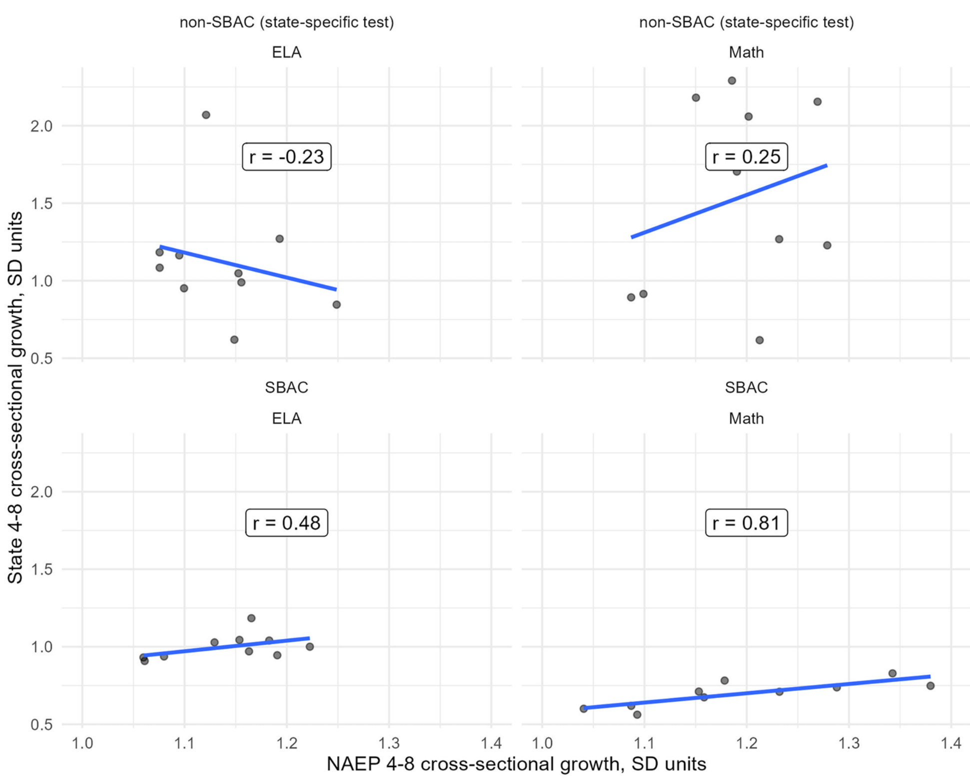

Figure 7. Association between grade 4–8 growth on NAEP and state tests.

are much more tightly clustered than growth estimates on state-specific scales. Second, although R^2 is higher for SBAC states in both subjects, there are also potentially interesting differences across subjects. NAEP results appear to track state test results more closely in math than in reading/ELA. One possible explanation for this would be if the NAEP math framework overlaps more with state math standards than the NAEP reading framework does with state ELA standards, but full exploration of this issue is beyond the scope of this article.

Discussion

Changing standards and psychometric/design issues in vertical scaling bear directly upon grade-to-grade growth magnitudes, the amount that a student must grow to progress from one grade's mean score to the next. When using these magnitudes as effect size benchmarks, it is important to align the benchmarks themselves with the outcome measure in one's study as closely as possible (Bloom et al., 2008), and this study provides benchmarks for grade 3-8 math and ELA based on 2019 state achievement tests. At the same time, this study's exploration of the extent to which growth across different state student populations is far more variable when states use different scales calls into question the circumstances under which the use of growth benchmarks based on an average computed across these different scales is advisable. I conclude this manuscript by considering this issue.

When These Growth Benchmarks Are Useful…

There can be substantial value in growth benchmarks when the outcome measure is not vertically scaled. For example, Kogan & Lavertu (2022) assessed the impact of the COVID-19 pandemic on student learning in Ohio, a state that does not use a vertical scale. There are many ways to try to contextualize an effect size on a non-vertically scaled test that are purely norm-referenced, such as Student Growth Percentile analysis and value-added models (Ballou, 2009; Betebenner, 2009; Briggs & Domingue, 2013), but benchmarking against annual growth benchmarks has been shown to hold particular appeal for end users (Lortie-Forgues et al., 2021).

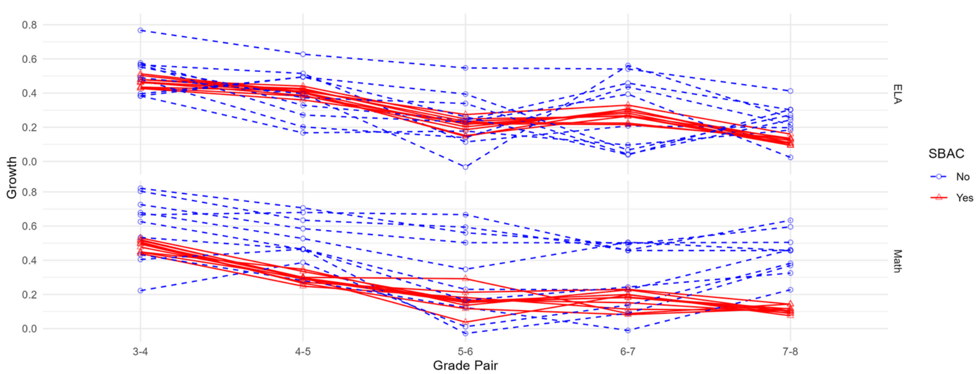

Figure 8. State-level growth on SBAC and non-SBAC scales.

The authors benchmarked their estimated causal effects against Bloom et al.'s benchmarks. I would suggest using the benchmarks in this study instead, as growth in 2019 on state tests is more relevant to a study using state test results from the immediate subsequent years than is growth on pre-NCLB tests. The suggestion to use benchmarks that average across vertical scales in this scenario is based on the fact that (1) we would like to know what growth *would* look like on the outcome measure if it *were* vertically scaled, but (2) this is impossible to know with certainty because of the influence of the process of vertical scale construction itself on the resulting growth estimates. By summarizing what growth looks like across 11 different scales constructed with 11 different processes measuring the same subjects at the same time point, the benchmarks provided in this article are a "best guess" at what growth on a non-vertically scaled state achievement test could look like if it were vertically scaled. That is, the means (the benchmarks themselves) represent the average result of estimating annual growth using 11 different scales, and the *SD*s around those means represent how variable those growth estimates could realistically be; the analysis based on comparisons with NAEP shows that much of this variability is a result of using different scales, not just differences in how much students truly learn. As emphasized throughout this article, focusing on the means alone masks the substantial variability in growth across scales. Researchers might consider, for example, benchmarking a causal effect on a non-vertically scaled state achievement test against the mean for the relevant grade and subject presented in this article, plus or minus e.g. one standard deviation. Specific procedures will depend upon the extent to which the researcher is substantively interpreting the magnitude of their causal effect in months or years of learning. The greater the reliance on this framing, the more sensitivity analysis the researcher should conduct, especially because "years of learning" framing of effect sizes has been shown to lead teachers to overestimate an intervention's effectiveness (Lortie-Forgues et al., 2021).

Ultimately, there is no single "right" way to use effect size benchmarks. Researcher subjectivity is always part of the research process, from power analysis and meta-analysis design through comparison of causal effects. In all cases, it is critical to recognize both the value and the limitations of effect size benchmarks, whether those introduced in this study, existing benchmarks to which I compared them, or another set entirely (e.g. Scammacca et al., 2015; Tanner-Smith et al., 2018).

...and When They Are Not

This article was motivated in part by a concern that the comparability of growth magnitudes across different achievement scales within the same subject is limited, based on the number of different scale design and calibration decisions that go into creating a vertical scale. The exploratory portion of this article provided descriptive evidence in line with the concern that growth effect sizes on different scales vary more than underlying learning, and vary as a function of how the scale was constructed. A consequence of this is that the benchmarks I provided in this article may be a worse estimate of annual growth on any single vertical scale than simply calculating "years of learning" based on prior-year results on that scale alone.

It follows that when researchers are conducting studies where the outcome is vertically scaled, it makes more sense to benchmark any causal effect estimates against prior growth only on that scale. The state-specific means, standard deviations and sample sizes used in this study are available in the online supplementary materials for this purpose. Researchers should also be cognizant of the population whose scores are the basis for their benchmarks (Baird & Pane, 2019). For example, in the context of post-COVID intervention studies, the benchmarks in this study might be framed as a "typical pre-COVID year of learning," but a post-COVID year of learning could differ from this study's benchmarks (an area for future research). Many states make scale score data publicly available in data portals and/or are responsive to data requests, so acquiring the requisite descriptive statistics to tailor vertical scale-specific benchmarks should be feasible for most researchers using a vertically scaled test score as an outcome measure. That said, there were also several states that were unresponsive to my efforts to gather the basic descriptive information that I used in this study. Researchers using scores from those tests as outcomes likely cannot construct their own benchmarks. The results of Analysis 3 of this study suggests that researchers in this situation should attempt to find the technical manual for their outcome measure to see if they can at least figure out which linking design was used to construct the scale, given how much variance in growth was explained when linking design was included as a predictor. They can then create their own benchmarks across all scales in this study that use that linking design; this information is contained in the online supplementary materials.

It is also important to note that there are numerous vertically scaled tests beyond state achievement tests, and that they warrant their own benchmarks. Interim assessments in math and reading such as MAP Growth (NWEA, 2019), i-Ready Diagnostic (Curriculum Associates, 2019), and STAR (Renaissance Learning, 2015) are growing in popularity as outcome measures. All three are vertically scaled, and researchers could benchmark causal effects on these outcomes against outcome-specific growth estimates. Additionally, these assessments can be used to track growth at a finer grain size than annual summative tests, and spring-to-spring growth benchmarks may not be the ideal benchmark when growth is being tracked on a shorter timeline. This introduces a new set of challenges and opportunities for finer-grained growth models and benchmarks; see Thum and Kuhfeld (2020, p. 66-73) for a discussion of the unique considerations in the interim context beyond the challenges of annual growth benchmarking discussed in this article.

Conclusion

This study presents updated growth benchmarks for state achievement tests of math and reading in grades 3–8, while also exploring the extent to which and reasons why growth varies across scales. The application of empirical benchmarks for growth is not straightforward. Despite criticism of their application to date (Baird & Pane, 2019), it is also apparent that when communicating causal effects, they have an appeal not shared by other methods of translating effect sizes to more familiar terms (Lortie-Forgues et al., 2021). Given this demand for translation of causal effects into potentially fraught terms, researchers should do their best to match their benchmarks to their context while recognizing that there will always be some degree of uncertainty in research involving unobservable constructs, and effect size benchmarks can be a tool to manage this uncertainty–or to exacerbate it.

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Open Research Statements

Study and Analysis Plan Registration

There is no registration associated with this study.

Data, Code, and Materials Transparency

The data supporting the findings of this study are available within the article's supplementary materials, though they are not available in a public repository. The code associated with this study is not publicly available.

Design and Analysis Reporting Guidelines

Not applicable.

Transparency Declaration

The lead author (the manuscript's guarantor) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

Replication Statement

This manuscript reports an original study.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

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